

Field Requirements

Primary project source: ProofRoad v2, SIH 2026 PS 26007, uploaded by the user. The document describes ProofRoad as an engineering proposal and explicitly says it is not a safety certification. Page references below refer to the uploaded ProofRoad v2 document.

1. Functional requirements

FR-01: Detect relevant obstacles ahead using radar.

FR-02: Maintain a road-constrained hazard representation so objects on another branch are not treated as hazards on the current route.

FR-03: Calculate proven-clear distance conservatively.

FR-04: Calculate needed stopping distance from current speed, reaction time, braking value and buffer.

FR-05: Advise a safe speed and stop when proven distance is insufficient.

FR-06: Share age/sequence/signature-tagged hazard messages between trucks.

FR-07: Support fail-closed blind-curve gates.

FR-08: Continue local safety behavior when network, camera or assist computer fails.

FR-09: Provide replayable decision logs and near-miss records.

FR-10: Provide an operator interface requiring only a brief glance.

2. Performance requirements to measure

Metric	Prototype target / requirement	Status
Radar detection	Measure experimentally; do not invent mine-grade range	To validate
Safe-speed computation	Deterministic and replayable	Architecture defined
Alert behavior	One alert per hazard; escalation repeats	Defined
Network failure	No loss of local radar safety loop	Designed
Message freshness	Age counter + sequence number	Designed
Gate uncertainty	Unknown/uncertain blocks entry	Designed
Logging	10 s before + 20 s after warning	Defined
Fleet effect	Compare against crawl-in-fog baseline	Simulation planned

3. Hardware requirements

Two model vehicles; radar or clearly labelled stand-in; safety MCU; IMU; wheel encoder; radio; screen; buzzer; optional vibration motor and LED bar.

Sensor add-on per truck: thermal array module, NoIR camera and 850 nm lamp. ProofRoad gives a planning add-on estimate of about ■5,000–■15,000 per truck for these modules and advises verification of price and lead time (p. 11).

Fog posts: scatter/visibility sensor plus temperature and humidity, with LoRa; blind-curve gates: lamp and break-beam sensor; laptop for dashboard/replay; mist source and known-distance targets.

4. Deployment requirements

Mine-specific road map and route checkpoints.

Truck-specific braking tests under representative load, slope and surface.

Environmental calibration for fog, dust, rain and thermal conditions.

Radio range tests in pit geometry.

Cybersecurity key provisioning and revocation for production.

Formal safety analysis before any automatic vehicle intervention.